

SEAT No. : _____



No. of Pages : 2

[67]

SARDAR PATEL UNIVERSITY

Master of Science (Information Technology)

Semester – III

PS03EINT56 : Software Testing

Date : 15/04/2024 (Monday)

Time: 02:00 PM to 5:00 PM

Max. Marks: 70

Q-1 Pick up the most appropriate answer from the given alternatives and write it in your answer book. (8)

- (i) Acceptance testing is done by -
[A] Customers [B] Testers [C] Developers [D] Programmers
- (ii) RTM stands for -
[A] Requirement Traceability Matrix [B] Requirement Trending Matrix
[C] Requirement Testing Matrix [D] Requirement Tracing Matrix
- (iii) For a function of n variables, boundary value analysis generates _____ test cases.
[A] $4n+1$ [B] $6n+1$ [C] $8n+1$ [D] $4n-1$
- (iv) Testing a software primarily focus on
[A] Verification only [B] Validation only
[C] Both [A] and [B] [D] None of these
- (v) A set of inputs, execution of preconditions and expected outcomes is known as _____.
[A] Test Case [B] Test Plan [C] Test Document [D] Test Suite
- (vi) Which of the followings is not a part of a test design documents?
[A] Test Log [B] Test Design Specification
[C] Test Plan [D] Test Case Specification
- (vii) Software testing with real data in real environment is known as _____.
[A] Beta Testing [B] Alpha Testing
[C] Regression Testing [D] None of these
- (viii) Which of the following is true for Selenium?
[A] Selenium supports Java only
[B] Selenium supports Ruby only
[C] Selenium supports Python only
[D] Selenium supports Java, Ruby and Python.

Q-2 Attempt the following: (ANY SEVEN)

(14)

- i. Define : Software. Write the role of software in any business.
- ii. Define the terms: 'Failure' and 'Error'.
- iii. Differentiate: Verification and Validation.
- iv. Differentiate: White-box Testing and Functional Testing.

(Contd../2)

- v. What is statement testing? Why we do it?
- vi. Write down the uses of effective test cases.
- vii. What is Test Case? Draw basic Test Case Template.
- viii. What is an open-source testing tool? Give name of any four popular open-source testing tools.
- ix. Write merits and demerits of testing tools.

- Q-3 (a)** Explain STLC in detail. (6)
(b) Explain various levels of software testing. (6)

OR

- (b)** Explain - principles of software testing (6)

- Q-4 (a)** What is black-box testing? Explain any two black-box testing techniques. (6)
(b) Explain – 'Branch Coverage' and 'Path Coverage' testing techniques (6)

OR

- (b)** What is white-box testing? Explain any two white-box testing techniques. (6)

- Q-5 (a)** Explain - test case parameters. (6)
(b) Explain - automated test data generation. (6)

OR

- (b)** Explain test adequacy criteria and test selection mechanism. (6)

- Q-6 (a)** Explain – 'Selenium as a Testing Tool. (6)
(b) Describe the role of testing tools with its advantages and disadvantages. (6)

OR

- (b)** Write a note on manual testing and automated testing with their advantages and disadvantages. (6)

----- ***** -----